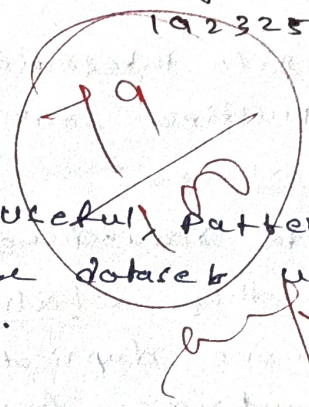




1) a) Data Mining, ADD & Preprocessing

Data Mining: The process of discovering useful patterns, correlations, & knowledge from large datasets using statistical, mathematical & ML tech.

Two basic data mining tasks:

1) Classification: Assign data to predefined categories
Ex: Spam vs Not Spam.

2) Clustering: Group similar data points without predefined labels.

Ex: Customer Segmentation

Steps involved in ADD

- 1) Data Selection
- 2) Data Preprocessing
- 3) Data Mining

Role of data processing

Data is usually noisy, incomplete & inconsistent. Preprocessing improves data quality.

b) Data cleaning

The process of detecting and correcting inaccurate, incomplete, or irrelevant data to improve quality.

1) Missing values

2) Noisy data

2) a) Data Integration: It is the process of combining data from multiple sources into a single consistent dataset.

Challenges:

Data redundancy and data inconsistency.

Handling challenges:

- * Remove duplicate records to deduce redundancy.
- * Standardize formats & resolve conflicting values to eliminate inconsistency.

b) Covariance Analysis: It measures how two variables change together. It indicates whether the variables increase or decrease together.

Types:

- 1) Positive covariance.
- 2) Negative covariance.

3) Suppose two stocks A & B have the values:

$A = (2, 3, 4, 5, 6)$ $B = (5, 8, 10, 11, 12)$

The variance of two stocks is +ve (this is a positive covariance indicates that both stocks tend to rise & fall together. Therefore they are positively related).

- 1) Databases: Store large amount of data.
- 2) Data base server: Retrieve & manages data for mining.
- 3) Data mining engine: Performs tasks like classification, clustering & analysis.
- 4) Pattern evolution module: Identifies useful & interesting patterns.
- 5) Knowledge base: Store domain knowledge.

4) Covariance for
 $x = \{1, 5, 6, 7, 9\}$ $y = \{0, 3, 7, 5, 6\}$
 mean of $x = 6.2$
 mean of $y = 5.8$
 covariance = 0.38

* The covariance is 0.38, indicating a slight negative relationship b/w x & y

6) Major steps in data preprocessing

- 1) Data cleaning: Remove missing values and noise
- 2) Data integration: Combine data from multiple source.
- 3) Data transformation: Convert data into a suitable format
- 4) Data discretization: Convert continuous values into intervals.

5) Perform equal-frequency binning:

Given data: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 40, 45, 46, 52, 70.

Divide into 3 bins:

Bin 1: 13, 15, 16, 16, 19, 20, 20, 21, 22

Bin 2: 22, 25, 25, 25, 25, 30, 33, 33, 35

Bin 3: 35, 35, 35, 36, 40, 45, 46, 52, 70

Equal frequency binning place an equal no. of data value in each bin, helping reduce noise.

b) Data discretization: It is the process of converting continuous data into a finite no. of intervals.

Importance:

- * Reduce data size
- * Remove noise
- * Improves mining efficiency
- * Simplifies analysis.

Comparison:

Equal width Binning

Equal interval size

Simple to implement

Equal-frequency Binning

Equal no. of records

Better for skewed data.

